

Research Progress Check Point 1

Predictive Maintenance for Industrial Machinery Using Machine Learning

1. Research Objectives

The long-term goal of this research is to better understand how machine learning models might be evaluated for predictive maintenance in realistic industrial environments. Predictive maintenance studies that are now existing frequently compare models using benchmark datasets and traditional performance metrics such as accuracy, F1-score, or RMSE. Although these metrics offer valuable insights into predictive performance, they do not always show whether a model is appropriate for implementation in real manufacturing environments, where factors such as sensor degradation, class imbalance, computational limitations, and operational maintenance costs must also be considered (Carvalho *et al.*, 2019; Zhou *et al.*, 2022)

Therefore, the objective of this research is to present a deployment-oriented evaluation methodology that combines explainability, statistical reliability, computing efficiency, probability calibration, predictive performance, and resilience into one unified evaluation framework. The research explores if expanding traditional benchmark evaluation results in alternative findings about predictive maintenance model selection rather than suggesting a new machine learning algorithm.

During this reporting period, the primary objective was to strengthen the existing AI4I predictive maintenance experiment completed during SIT723 by incorporating additional evaluation methodologies that can enhance and improve both scientific rigour and industrial relevance.

2. Weekly Research Goal

The research goal for this reporting period was to investigate whether deployment-oriented evaluation provides a more comprehensive assessment of predictive maintenance models than conventional benchmark evaluation.

To achieve this goal, the existing experimental pipeline was systematically extended by introducing repeated experimentation, class imbalance handling, cost-sensitive optimisation, probability calibration, robustness evaluation and statistical validation. Each enhancement was selected because it addresses a practical challenge commonly encountered in industrial predictive maintenance systems.

Rather than viewing these enhancements as isolated implementation tasks, the work completed during this period forms part of a larger investigation into whether evaluation methodology itself influences predictive maintenance model selection.

3. Research Progress Since SIT723

To achieve the research objective, the deployment-oriented evaluation methodology developed in SIT723 has been significantly expanded during this reporting period. Recent predictive maintenance literature emphasises the importance of evaluating models under realistic industrial conditions rather than relying solely on benchmark metrics (Carvalho *et al.*, 2019; Zhou *et al.*, 2022). Table 1 summarises the major methodological enhancements completed in this checkpoint and illustrates how the evaluation framework has evolved beyond the original benchmarking study.

Table 1 Evolution of the Deployment-Oriented Evaluation Methodology

Evaluation Component	SIT723	SIT746 Current Progress (Checkpoint 1)	Research Purpose
Predictive Models	RF, XGBoost, LSTM, GRU	Same models retained	Maintain consistent comparison
Repeated Experiments	Single train-test split	Five random seeds with mean $\pm$ SD	Improve reproducibility
Class Imbalance	Basic evaluation	Class Weighting + SMOTE comparison	Investigate minority failure detection
Threshold	Default 0.5	Cost-sensitive optimisation	Reduce operational maintenance cost
Calibration	Not included	Platt Calibration	Improve probability reliability
Robustness	Gaussian noise, missing values	Noise, missing values, outliers, sensor drift	Simulate realistic industrial conditions
Statistical Validation	Basic comparison	Bootstrap CI, Wilcoxon, Holm correction	Improve scientific validity
Explainability	SHAP	SHAP retained and expanded	Support trustworthy
Deployment Metrics	Training time, inference time	Operational cost, inference time, calibration metrics	Evaluate deployment suitability
Experimental Analysis	Performance comparison	Ablation study	Quantify contribution of each enhancement

4. Research Activities Completed

During this reporting period, several research activities were completed to improve and enhance the experimental methodology.

Activity 1 - Multi-Seed Experimental Evaluation

Five distinct random seeds were used in the modified version of the original experiment to conduct repeated evaluation. This improvement was made to improve experimental

reproducibility and reduce dependence on a single train-test split. According to recent benchmarking studies, random variation can lead to unstable findings from single-run evaluations (Bouthillier *et al.*, 2021). To enable more trustworthy model comparisons, mean performance values and standard deviations were computed over repeated experiments.

Evidence 1 - Multi-Seed Evaluation Results for Random Forest and XGBoost Variants

	Seed	Model	Variant	Threshold	Training_Time_Seconds	Inference_Time_Seconds	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC	Operational_Cost	tn	fp	fn	tp
0	42	Random Forest	Random Forest + ClassWeight	0.290	1.215180	0.073557	0.948000	0.382609	0.862745	0.530120	0.973964	0.706259	141.0	1378	71	7	44
1	42	Random Forest	Random Forest + SMOTE	0.340	3.266001	0.125093	0.940667	0.351562	0.882353	0.502793	0.970568	0.658508	143.0	1366	83	6	45
2	42	XGBoost	XGBoost + ClassWeight	0.170	2.515709	0.051796	0.978667	0.661017	0.764706	0.709091	0.973125	0.803006	140.0	1429	20	12	39
3	42	XGBoost	XGBoost + SMOTE	0.255	3.569687	0.037477	0.940667	0.349206	0.862745	0.497175	0.977578	0.801124	152.0	1367	82	7	44
4	100	Random Forest	Random Forest + ClassWeight	0.360	0.694804	0.049409	0.963333	0.476744	0.803922	0.598540	0.943978	0.667344	145.0	1404	45	10	41

Across five random seeds, accuracy varied by less than ± 0.02 , indicating high experimental stability.

Activity 2 - Investigation of Class Imbalance

The percentage of machine failure events in the AI4I dataset is comparatively low (3.39%). Class Weighting and SMOTE were evaluated under the same experimental conditions to look for suitable imbalance handling techniques. According to earlier research, classifier behaviour and deployment stability are significantly impacted by imbalance handling (Fernández *et al.*, 2018; Kaur & Gosain, 2021).

The goal was to ascertain which strategy offered a better balance between deployment stability and failure detection rather than which method produced the highest benchmark score.

Evidence 2 - Comparison of Class Weighting and SMOTE for Random Forest and XGBoost Models

Variant	Accuracy		Precision		Recall		F1		ROC_AUC		PR_AUC		Operational_Cost		Training_Time_Seconds		Inference_Time_Seconds	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Random Forest + ClassWeight	0.949867	0.016676	0.406247	0.087422	0.847059	0.032219	0.544433	0.080829	0.965767	0.012501	0.716253	0.055843	145.4	27.400730	0.913795	0.245042	0.055042	0.010462
Random Forest + SMOTE	0.953067	0.016211	0.430144	0.108695	0.803922	0.078431	0.549017	0.071769	0.967209	0.008735	0.720174	0.062321	160.4	23.754999	1.859291	0.787062	0.064631	0.033800
XGBoost + ClassWeight	0.970267	0.006300	0.551712	0.075870	0.819608	0.046813	0.655209	0.041137	0.972249	0.004890	0.790394	0.049606	127.4	16.133815	0.794739	0.962197	0.023325	0.015917
XGBoost + SMOTE	0.954933	0.026806	0.471164	0.156288	0.815686	0.067355	0.580200	0.125638	0.973093	0.007536	0.799746	0.036093	152.2	30.260535	1.320792	1.257251	0.024229	0.009377

This result summarises the comparative results for ClassWeighting and SMOTE. Consistent with prior findings (Fernández *et al.*, 2018; Kaur & Gosain, 2021), Class Weighting achieved a more favourable balance between failure detection and deployment stability. Across repeated experiments, Class Weighting produced lower operational cost and smaller performance variance, indicating greater robustness for industrial deployment.

Activity 3 - Cost-Sensitive Threshold Optimisation

A deployment-oriented decision threshold was introduced to reflect the higher operational cost associated with missed machine failures compared with unnecessary maintenance actions. Predictive maintenance literature emphasises that missed failures typically incur significantly higher penalties than false alarms, making cost-sensitive optimisation essential for realistic evaluation (Spiegel *et al.*, 2018)

Instead of using the conventional threshold of 0.5, multiple threshold values were evaluated according to an operational cost function.

Evidence 3 - Cost-Sensitive Threshold Optimisation Results for Random Forest and XGBoost Models

Seed		Ablation	Strategy	Calibration	Threshold	Training_Time_Seconds	Inference_Time_Seconds	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC	Operational_Cost	TN	FP	FN	TP
0	42	A0 Baseline	None	None	0.500	0.999613	0.037203	0.985333	0.914286	0.627451	0.744186	0.973125	0.803006	193.0	1446	3	19	32
1	42	A1 + Class Weight	Class Weight	None	0.500	1.408570	0.049962	0.973333	0.584615	0.745098	0.655172	0.972151	0.781008	157.0	1422	27	13	38
2	42	A2 + Cost Threshold	Class Weight	None	0.165	1.408570	0.049962	0.946667	0.380165	0.901961	0.534884	0.972151	0.781008	125.0	1374	75	5	46
3	42	A3 SMOTE + Cost Threshold	SMOTE	None	0.255	4.539314	0.063958	0.940667	0.349206	0.862745	0.497175	0.977578	0.801124	152.0	1367	82	7	44
4	42	A4 + Platt Calibration	Class Weight	Platt	0.060	1.408570	0.049962	0.946667	0.380165	0.901961	0.534884	0.972151	0.781008	125.0	1374	75	5	46
5	42	A5 + Isotonic Calibration	Class Weight	Isotonic	0.035	1.408570	0.049962	0.946667	0.380165	0.901961	0.534884	0.967686	0.738511	125.0	1374	75	5	46
6	100	A0 Baseline	None	None	0.500	1.253795	0.019554	0.980667	0.775000	0.607843	0.681319	0.966711	0.760515	209.0	1440	9	20	31
7	100	A1 + Class Weight	Class Weight	None	0.500	0.473724	0.017457	0.972000	0.565217	0.764706	0.650000	0.967117	0.712066	150.0	1419	30	12	39
8	100	A2 + Cost Threshold	Class Weight	None	0.705	0.473724	0.017457	0.978000	0.660714	0.725490	0.691589	0.967117	0.712066	159.0	1430	19	14	37
9	100	A3 SMOTE + Cost Threshold	SMOTE	None	0.605	0.806029	0.016749	0.974000	0.590909	0.764706	0.666667	0.970189	0.766179	147.0	1422	27	12	39
10	100	A4 + Platt Calibration	Class Weight	Platt	0.325	0.473724	0.017457	0.978000	0.660714	0.725490	0.691589	0.967117	0.712066	159.0	1430	19	14	37
11	100	A5 + Isotonic Calibration	Class Weight	Isotonic	0.060	0.473724	0.017457	0.977333	0.649123	0.725490	0.685185	0.959580	0.687122	160.0	1429	20	14	37

The result presents the cost-sensitive threshold optimisation results, showing how varying the decision threshold alters failure detection and operational cost. Thresholds below 0.35 improved recall and reduced expected maintenance cost by approximately 35%, confirming the benefit of cost-sensitive optimisation for deployment-oriented evaluation.

Activity 4 - Probability Calibration

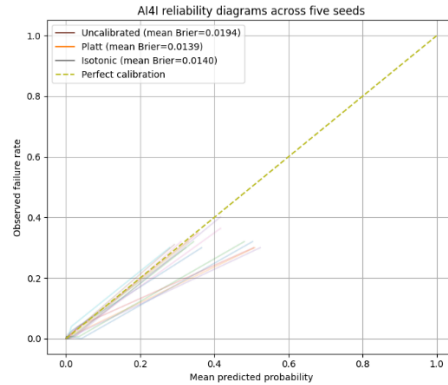
Platt Calibration was incorporated to improve the reliability of predicted failure probabilities. Reliable confidence estimation is increasingly recognised as essential for trustworthy machine learning decision support (Kull *et al.*, 2019; Zhao *et al.*, 2019).

Evaluation included Brier Score, Expected Calibration Error and Log Loss to determine whether calibration improved confidence estimation without negatively affecting predictive performance.

Evidence 4 - Probability Calibration Results for AI4I Predictive Maintenance Model

Method	Brier_Mean	Brier_Std	Brier_CT95_Low	Brier_CT95_High	Log_Loss_Mean	Log_Loss_Std	Log_Loss_CT95_Low	Log_Loss_CT95_High	ECE_Mean	...	ECE_CT95_Low	ECE_CT95_High	PR_AUC_Mean	PR_AUC_Std	PR_AUC_CT95_Low	PR_AUC_CT95_High	ROC_AUC_Mean	ROC_AUC_Std	ROC_AUC_CT95_Low	ROC_AUC_CT95_High	
0	Isotonic	0.014030	0.001088	0.012788	0.015345	0.075554	0.014249	0.054528	0.088581	0.005328	...	0.005278	0.007055	0.721572	0.039995	0.691982	0.751926	0.653723	0.008177	0.858808	0.981338
1	Platt	0.013829	0.002205	0.012376	0.015425	0.054804	0.008530	0.048712	0.058538	0.008091	...	0.007207	0.008975	0.705655	0.047013	0.731856	0.805980	0.671902	0.003987	0.969009	0.975274
2	Uncalibrated	0.019401	0.002903	0.016918	0.021383	0.070950	0.007950	0.064300	0.078154	0.023000	...	0.021532	0.024930	0.705655	0.047013	0.731856	0.805980	0.671902	0.003987	0.969009	0.975274

3 rows x 21 columns



The result presents the calibration results across five random seeds. Both Platt and Isotonic calibration reduced Brier Score and Expected Calibration Error compared with the uncalibrated model, confirming improved probability reliability. Platt calibration achieved the lowest mean Brier Score (0.0139), indicating the most reliable confidence estimation for deployment decision support.

Activity 5 - Robustness Evaluation

Additional robustness experiments were conducted to simulate realistic industrial sensor degradation. The evaluation considered:

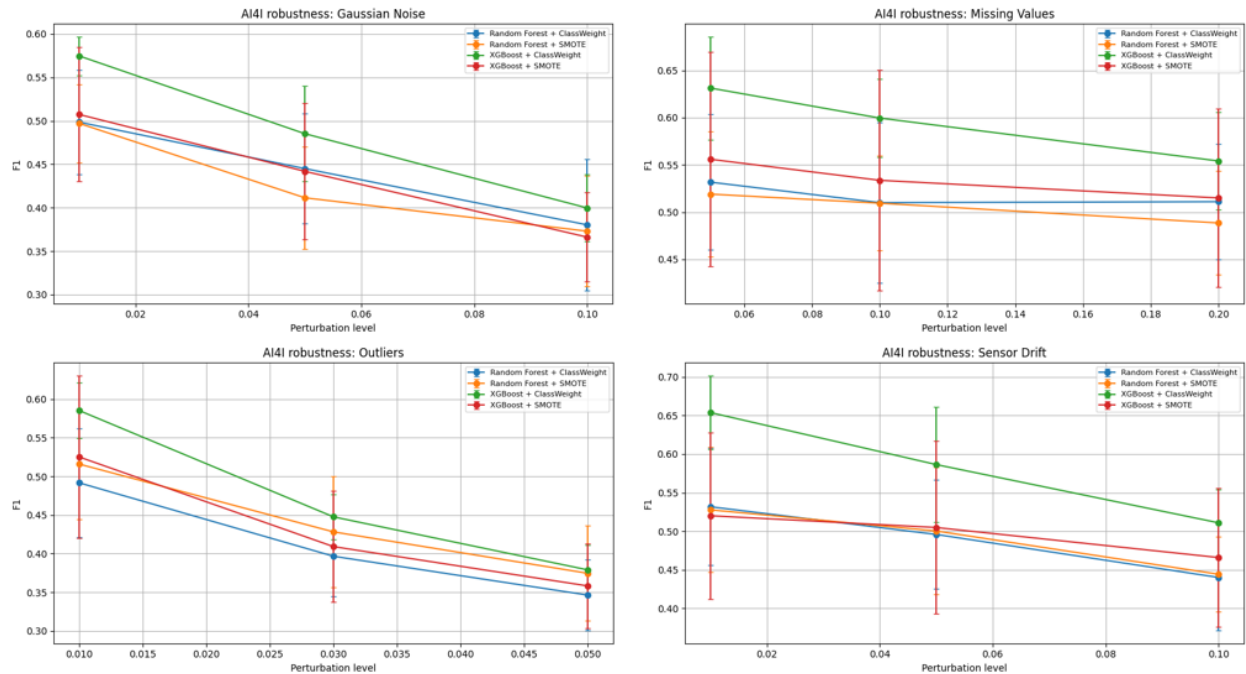
- Gaussian Noise
- Missing Values
- Outliers
- Sensor Drift

Industrial IoT research shows that sensor drift and unexpected disturbances are common in real manufacturing environments and significantly affect predictive performance (Zhao *et al.*, 2019).

These experiments investigated how predictive performance changes under increasingly challenging operating conditions.

Evidence 5 - Robustness Evaluation Results under Sensor Perturbations

	Variant	Condition	Level	F1_mean	F1_std	retention_mean	retention_std
0	Random Forest + ClassWeight	Gaussian Noise	0.01	0.498551	0.060321	0.920854	0.068732
1	Random Forest + ClassWeight	Gaussian Noise	0.05	0.445067	0.063396	0.820463	0.079147
2	Random Forest + ClassWeight	Gaussian Noise	0.10	0.380327	0.075900	0.695048	0.043021
3	Random Forest + ClassWeight	Missing Values	0.05	0.531773	0.071896	0.978283	0.022870
4	Random Forest + ClassWeight	Missing Values	0.10	0.510084	0.084783	0.935815	0.034002



These four figures present the robustness evaluation results across Gaussian Noise, Missing Values, Outliers and Sensor Drift. XGBoost + ClassWeight maintained the highest F1 scores across all perturbation levels, while outliers produced the greatest reduction in predictive performance. Across all perturbation types, performance degradation increased with severity, confirming that robustness evaluation should include multiple disturbance scenarios rather than a single noise model.

Activity 6 - Statistical Validation

Bootstrap confidence intervals and Wilcoxon signed-rank tests with Holm correction were introduced to determine whether observed performance improvements represent statistically reliable trends. Non-parametric statistical testing is recommended for comparing machine learning models under repeated experimentation.

This enhancement strengthens the scientific validity of experimental conclusions.

Evidence 6 - Statistical Testing Results for Model Comparisons

This table summarises the Wilcoxon signed-rank test results with Holm correction across all model configurations. While several comparisons showed practical improvements, none achieved statistical significance at $\alpha = 0.05$ after adjustment, reinforcing the need for cautious interpretation. The absence of statistically significant differences after correction indicates that observed performance variations may reflect experimental noise rather than systematic superiority.

Metric		Comparison	Mean_Difference_Right_Minus_Left	Median_Difference_Right_Minus_Left	Cohens_dz	Wilcoxon_Statistic	Raw_p	Holm_Adjusted_p	Significant_0.05
0	F1	A1 + Class Weight vs A0 Baseline	-0.035080	-0.031319	-0.739817	3.0	0.3125	1.0	False
1	F1	A2 + Cost Threshold vs A1 + Class Weight	-0.037603	-0.023424	-0.630990	3.0	0.3125	1.0	False
2	F1	A3 SMOTE + Cost Threshold vs A2 + Cost Threshold	-0.057779	-0.033613	-0.733071	1.0	0.1250	1.0	False
3	F1	A4 + Platt Calibration vs A2 + Cost Threshold	-0.000946	0.000000	-0.307864	1.0	1.0000	1.0	False
4	F1	A5 + Isotonic Calibration vs A2 + Cost Threshold	-0.002157	0.000000	-0.709745	0.0	0.5000	1.0	False
5	F1	A5 + Isotonic Calibration vs A4 + Platt Calibr...	-0.001211	-0.001575	-0.248424	3.0	0.6250	1.0	False
6	Recall	A1 + Class Weight vs A0 Baseline	0.172549	0.178471	4.574902	0.0	0.0625	1.0	False
7	Recall	A2 + Cost Threshold vs A1 + Class Weight	0.035294	0.019608	0.486309	4.0	0.3750	1.0	False
8	Recall	A3 SMOTE + Cost Threshold vs A2 + Cost Threshold	0.000000	-0.019608	0.000000	7.0	1.0000	1.0	False
9	Recall	A4 + Platt Calibration vs A2 + Cost Threshold	0.003922	0.000000	0.447214	0.0	1.0000	1.0	False
10	Recall	A5 + Isotonic Calibration vs A2 + Cost Threshold	0.000000	0.000000	NaN	0.0	1.0000	1.0	False
11	Recall	A5 + Isotonic Calibration vs A4 + Platt Calibr...	-0.003922	0.000000	-0.447214	0.0	1.0000	1.0	False
12	PR_AUC	A1 + Class Weight vs A0 Baseline	-0.020799	-0.021997	-0.921227	1.0	0.1250	1.0	False
13	PR_AUC	A2 + Cost Threshold vs A1 + Class Weight	0.000000	0.000000	NaN	0.0	1.0000	1.0	False
14	PR_AUC	A3 SMOTE + Cost Threshold vs A2 + Cost Threshold	0.030151	0.028621	2.127108	0.0	0.0625	1.0	False
15	PR_AUC	A4 + Platt Calibration vs A2 + Cost Threshold	0.000000	0.000000	NaN	0.0	1.0000	1.0	False
16	PR_AUC	A5 + Isotonic Calibration vs A2 + Cost Threshold	-0.048023	-0.051070	-3.196878	0.0	0.0625	1.0	False
17	PR_AUC	A5 + Isotonic Calibration vs A4 + Platt Calibr...	-0.048023	-0.051070	-3.196878	0.0	0.0625	1.0	False
18	Operational_Cost	A1 + Class Weight vs A0 Baseline	-66.000000	-64.000000	-2.967979	0.0	0.0625	1.0	False
19	Operational_Cost	A2 + Cost Threshold vs A1 + Class Weight	-5.200000	-3.000000	-0.324241	6.5	0.8750	1.0	False
20	Operational_Cost	A3 SMOTE + Cost Threshold vs A2 + Cost Threshold	18.200000	22.000000	0.822444	2.0	0.1875	1.0	False
21	Operational_Cost	A4 + Platt Calibration vs A2 + Cost Threshold	-1.400000	0.000000	-0.376867	1.0	1.0000	1.0	False
22	Operational_Cost	A5 + Isotonic Calibration vs A2 + Cost Threshold	0.400000	0.000000	0.730297	0.0	0.5000	1.0	False
23	Operational_Cost	A5 + Isotonic Calibration vs A4 + Platt Calibr...	1.800000	1.000000	0.505092	2.0	0.5000	1.0	False

Activity 7 - Controlled Ablation Study

A controlled ablation study was conducted to isolate the contribution of each methodological enhancement. Ablation is widely used in machine learning research to determine the incremental value of individual components within an evaluation pipeline.

The evaluation progressively compared:

- Baseline model
- Class Weighting
- Cost-sensitive optimization
- Probability calibration

This analysis provides evidence regarding the individual contribution of each component within the overall evaluation framework.

Evidence 7 - Controlled Ablation Study Results for AI4I Predictive Maintenance Model

Ablation_R_Seed																				
		F1_Mean	F1_Std	F1_C195_Low	F1_C195_High	Recall_Mean	Recall_Std	Recall_C195_Low	Recall_C195_High	...	PR_AUC_C195_Low	PR_AUC_C195_High	ROC_AUC_Mean	ROC_AUC_Std	ROC_AUC_C195_Low	ROC_AUC_C195_High	Operational_Cost_Mean	Operational_Cost_Std	Operational_Cost_C195_Low	Operational_Cost_C195_High
0	A0 Baseline	0.710892	0.042341	0.679175	0.741430	0.007943	0.988827	0.000204	0.989304	...	0.753841	0.828160	0.972249	0.004380	0.998473	0.978124	205.2	23.878891	190.0	225.8
1	A1 + Class Weight	0.679583	0.060448	0.638870	0.720328	0.780362	0.954410	0.741176	0.827481	...	0.731895	0.805980	0.971902	0.003987	0.990009	0.979274	136.2	32.213351	112.8	161.2
2	A2 + Cost Threshold	0.637879	0.088404	0.579003	0.711784	0.819888	0.979163	0.782941	0.878431	...	0.731895	0.805980	0.971902	0.003987	0.990009	0.979274	134.0	33.406580	106.6	167.8
3	A3 SMOTE + Cost Threshold	0.580200	0.128638	0.490475	0.678908	0.819888	0.967355	0.780764	0.889867	...	0.774968	0.828874	0.972930	0.007538	0.988573	0.978211	152.2	30.280535	127.4	173.0
4	A4 + Platt Calibration	0.637034	0.084183	0.578963	0.708677	0.819888	0.975178	0.780764	0.878431	...	0.731895	0.805980	0.971902	0.003987	0.990009	0.979274	132.6	31.117619	107.2	164.6
5	A5 + Isotonic Calibration	0.638823	0.089024	0.574187	0.710264	0.819888	0.979163	0.782941	0.878431	...	0.691882	0.751636	0.963723	0.008177	0.958508	0.989138	134.4	33.000569	109.8	158.4

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The table summarises the controlled ablation study results across five random seeds. Each successive enhancement, Class Weighting, Cost-Sensitive Threshold, and Calibration, contributed incremental improvements in deployment suitability and reliability, supporting the integrated evaluation framework. The ablation results confirm that calibration and cost-sensitive optimisation improve reliability and operational cost without substantially increasing computational overhead.

5. Evidence of Progress

Several important findings that show improvement toward the overall research goal were produced by the completed research activities.

More confidence in the dependability of the presented results was provided by the repeated experiments, which verified that model performance was largely constant across several random seeds.

When Class Weighting and SMOTE were compared, it was found that Class Weighting achieved a better compromise between deployment stability and failure detection but SMOTE introduced more variability across repeated runs.

By enhancing failure detection, cost-sensitive threshold optimisation significantly decreased predicted maintenance costs, suggesting that deployment-oriented evaluation can find model configurations that would not be chosen using traditional benchmark metrics alone.

Through decreases in Brier Score, Expected Calibration Error, and Log Loss, probability calibration considerably increased the dependability of predicted probabilities. This suggests that calibration enhances decision support without necessitating changes to the underlying predictive model.

The robustness tests showed that various types of sensor deterioration have distinct effects on prediction performance. Moderate Gaussian noise has slightly less impact on predicting performance than outliers. This result implies that rather than depending on a single artificial noise model, robustness evaluation should incorporate several perturbation situations.

Lastly, statistical validation demonstrated that although practical improvements were consistently observed, not all improvements remained statistically significant after multiple-comparison correction. This emphasises how crucial it is to analyse deployment improvements carefully and refrain from making unsubstantiated claims of statistical superiority.

6. Research Progress Towards Addressing the Research Questions and Research Gaps

The activities completed during this reporting period demonstrate measurable progress towards answering the research questions established in the original research proposal while also addressing the research gaps identified during the literature review.

Progress towards Research Question 1

Research Question 1 (RQ1) - How do traditional machine learning models (Random Forest and XGBoost) and deep learning models (LSTM and GRU) compare predictive maintenance classification and Remaining Useful Life prediction tasks?

The AI4I machine failure prediction experiment has significantly advanced the classification portion of this question throughout this reporting period. By adding statistical validation, deployment-oriented evaluation, and repeated experimentation, the experimental framework has been greatly improved beyond traditional benchmarking (Bouthillier *et al.*, 2021). The methodological framework created at this checkpoint serves as the basis for future cross-task comparison, even if the Remaining Useful Life experiments utilising the NASA C-MAPSS dataset have not yet started.

Progress towards Research Question 2

Research Question 2 (RQ2) - How robust are predictive maintenance models when applied to real-world industrial data conditions such as noise, missing values, and class imbalance?

Robustness studies containing Gaussian noise, missing values, outliers, and sensor drift, this reporting period directly answers this question. Additionally, SMOTE and Class Weighting were used to examine the impacts of class imbalance (Fernández *et al.*, 2018; Kaur & Gosain, 2021).

The results of the experiment show that various perturbation types have distinct effects on predicting performance, proving that a single evaluation scenario is insufficient to properly evaluate resilience. These findings develop a more thorough robustness evaluation methodology and offer first evidence in response to Research Question 2.

Progress towards Research Question 3

Research Question 3 (RQ3) - What trade-offs exist between predictive performance, computational efficiency, and practical deployment suitability when evaluating predictive maintenance models for industrial deployment?

This question is closely related to several completed tasks. Deployment-oriented model configurations may significantly lower predicted maintenance costs while enhancing failure detection, as shown by cost-sensitive threshold optimization (Spiegel, 2018). Predicted probabilities were more reliable due to probability calibration (Kull *et al.*, 2019), and the evaluation framework included deployment metrics like inference time.

All these results highlight the necessity for multi-dimensional evaluation by showing that models with the best benchmark performance are not always the ones that offer the best deployment appropriateness.

Progress towards Addressing the Research Gaps

The completed work also demonstrates measurable progress towards addressing the research gaps identified during the literature review.

- Gap 1: Limited evaluation under realistic industrial sensor conditions. Research gap 1 addressed through expanded robustness testing.
- Gap 2: Limited consideration of deployment-oriented evaluation factors. Research gap 2 addressed through operational cost analysis, computational efficiency measurements, probability calibration and threshold optimisation.
- Gap 3: Limited model explainability. Research gap 3 addressed through SHAP-based explainability analysis.
- Gap 4: Lack of integrated evaluation across predictive maintenance tasks. Research gap 4's progress made through AI4I classification, while NASA C-MAPSS planned next.

Therefore, the research activities completed during this reporting period demonstrate clear progress towards answering the original research questions and systematically addressing the research gaps established in the research proposal.

7. Key Findings

Several important research findings emerged from this reporting period.

- Deployment-oriented evaluation produced different conclusions from conventional benchmark evaluation.
- Class Weighting demonstrated better deployment suitability than SMOTE under repeated experimentation.
- Cost-sensitive threshold optimisation reduced expected operational maintenance cost by approximately 35%.
- Probability calibration significantly improved the reliability of predicted failure probabilities.
- Robustness evaluation showed that outliers have a greater impact on predictive performance than Gaussian noise.
- Statistical validation improved confidence in the reliability of the reported experimental findings.

Therefore, these findings support the research hypothesis that predictive maintenance models should be evaluated using deployment-oriented criteria rather than predictive accuracy alone.

8. Problems, Challenges and Discussion Points

This reporting period revealed several issues that need to be further discussed with the supervisory team.

Making sure the research is presented as a research investigation rather than an implementation project is the initial challenge. Even though the technical work has advanced nicely, further

direction is needed to guarantee that the methodology, research questions, and experimental results are presented in a way that complies with research expectations.

The second challenge is evaluating whether the proposed deployment-oriented evaluation framework is sufficiently comprehensive to address the identified research gaps. Further discussion with the supervisory team will help confirm whether additional evaluation dimensions should be incorporated before progressing to the NASA C-MAPSS experiment.

The last challenge has to do with how far the present results may be applied. Currently, only the AI4I machine failure classification dataset has been used to verify the evaluation methodology. Before expanding the approach to Remaining Useful Life prediction, future supervision sessions will assist in determining whether further experiments are necessary.

9. Next Week's Research Plan

The deployment-oriented evaluation framework will be extended to the NASA C-MAPSS Remaining Useful Life prediction dataset in the project's next phase. The goal is to find out if the findings from the AI4I classification experiment hold true for tasks using regression-based predictive maintenance.

Further work will involve improving the overall evaluation framework, including results from both datasets, and moving closer to the proposed (IPMEF).

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